Focused Hypothesis Report — F7A4N8 (Equus caballus CTDSP2): "enables kinase activity" (GO:0016301)

Executive Judgment

Verdict: REFUTED (kinase activity not supported), with the strongest alternative being a HAD-like protein phosphatase family.

The seed hypothesis is that horse protein **F7A4N8 enables kinase activity (GO:0016301)**. Independent sequence and orthology analysis refutes this:

1. **No kinase machinery.** The exact 174-aa sequence (SHA-256 verified) contains **none** of the defining eukaryotic protein-kinase catalytic motifs (Gly-rich loop GxGxxG, VAIK β3 lysine, HRD catalytic loop, DFG Mg-binding, APE). It has only 3 aspartates in 174 residues — incompatible with an ATP-phosphotransfer active site.
2. **The gene family is a phosphatase, not a kinase.** CTDSP2 belongs to the FCP/SCP small C-terminal-domain phosphatase family (HAD-like hydrolase superfamily). Its human ortholog O14595 is a reviewed **protein phosphatase** (active-site Asp107 forms a 4-aspartylphosphate intermediate). A phosphatase catalyzes **de**phosphorylation — mechanistically the opposite of a kinase.
3. **This particular horse model is truncated/aberrant.** F7A4N8 matches human CTDSP2 at ~96% identity over only the N-terminal ~84 residues, then diverges into a Ser/Arg-rich low-complexity C-terminus. It **lacks the entire FCP1-homology catalytic phosphatase domain** (human residues 97–255, DLDET active site). So this sequence supports neither kinase nor (on its own) phosphatase catalysis.

Most important caveat: F7A4N8 is a TrEMBL/Ensembl automatic prediction whose C-terminus appears to be an incomplete or mis-predicted gene model. Identity with the original prediction-time input was not independently establishable, but the supplied sequence was verified byte-for-byte (hash match).

Reproducible Methods & Provenance

- **Sequences (retrieved 2026-09-08 from UniProt REST `rest.uniprot.org/uniprotkb/{acc}.json`):**
- Target: **F7A4N8** (TrEMBL, *Equus caballus*), 174 aa, mass 19,012 Da; submitted name "CTD small phosphatase 2"; gene CTDSP2 (VGNC). SHA-256 = `94788163d6db3dbca3e1432762d7b2a1e8b57e1be2048c233c505a07a0b5042c` **(match confirmed)**.
- Human lead: **O14595** (Swiss-Prot), "Carboxy-terminal domain RNA polymerase II polypeptide A small phosphatase 2", 271 aa; FCP1-homology domain 97–255; active sites Asp107 (4-aspartylphosphate intermediate) and Asp109 (proton donor); Mg binding 218; keywords Hydrolase, Protein phosphatase, Magnesium, Nucleus; has PDB structure.
- **Motif scan** (regex against the verified sequence) — kinase hallmarks and FCP/SCP DxDxT/V motif.
- **Pairwise comparison** F7A4N8 vs O14595 (identity of shared prefix; divergence point; catalytic-domain presence).
- Computed outputs are reproduced in the Evidence Matrix and Mechanistic Scope below.

Independent cross-database checks (added Iteration 2): - **UniProt F7A4N8** has **no GO cross-references** and the word "kinase" appears **0 times** — the kinase claim is not database-derived. - **UniProt O14595** GO annotations: **GO:0008420** "RNA polymerase II CTD heptapeptide repeat phosphatase activity" with **IDA (direct assay)** evidence; **GO:0006470** protein dephosphorylation (IDA); GO:0005654 nucleoplasm (TAS). "kinase" appears 0 times; "phosphatase" 13 times. - **InterPro F7A4N8 → HTTP 204 (no domain matches at all)** — the horse model carries neither a kinase domain nor the phosphatase domain. - **InterPro O14595** → FCP1 homology domain (IPR004274), Dullard phosphatase domain (IPR011948), HAD-like superfamily (IPR036412), Pfam **PF03031 "NLI interacting factor-like phosphatase"**, SMART SM00577 "catalytic domain of ctd-like phosphatases", SFLD "RNA Pol CTD Phosphatase Like" — all phosphatase/HAD signatures, **zero kinase signatures**.

Structural check (added Iteration 3): AlphaFold DB model **AF-F7A4N8-F1 (v6, 2025-03-31)** — mean pLDDT **46.2 (very low)**; 66% of residues pLDDT <50, 98% <70; shared N-terminus (res 1–84) mean 51.4, divergent C-terminus (res 85–174) mean 41.4. **No high-confidence globular/catalytic fold**, consistent with the InterPro no-domain result.

Key computed values: - Kinase motifs GxGxxG / VAIK / HRD / DFG / APE: **all NONE**. - FCP/SCP catalytic motif DxDx[TV] / DLDET in F7A4N8: **NONE** ("DLDET" absent). - F7A4N8 vs O14595 identity over first 84 aa: **81/84 = 96.4%**; sequences track together until human ~res 84 (…CLQYQFYQ), then diverge. Human DLDET catalytic motif at residues 107–111 has **no counterpart** in the horse model. - F7A4N8 composition: Ser 16.7%, Arg 8.0%, Asp 1.7% (Asp at positions 14, 73, 147 only).

Evidence Matrix

#	Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context
1	This analysis (UniProt F7A4N8, 2026-09-08)	Computational (motif scan)	**Refutes**	F7A4N8 has kinase catalytic machinery	No GxGxxG/VAIK/ HRD/DFG/APE; only 3 Asp	Horse protein sequence
2	UniProt O14595 (Swiss-Prot)	Database (curated, structure-backed)	**Competing/ qualifies**	True MF of CTDSP2	Protein phosphatase; Asp107 4-aspartylphosphate intermediate; FCP1 domain; PDB structure	Human ortholog
3	This analysis (pairwise align)	Structural/ evolutionary	**Refutes/ qualifies**	Horse model encodes the catalytic domain	F7A4N8 = 96% id over N-term ~84 aa, then diverges; **lacks FCP1 catalytic domain / DLDET**	Horse vs human
4	Burkholder et al. 2018, P 30217818	Direct assay (kinetics + X-ray)	**Competing**	SCP family reaction direction	SCP1 **dephosphorylates** REST degron pSer-861 (phosphatase)	Human, in vitro + HEK293
5	Visvanathan et al. 2007, P 17403776	Mutant/ overexpression phenotype	**Competing**	SCP family is a phosphatase	"phosphatase SCP1 (small C-terminal domain phosphatase 1)" controls neurogenesis	Mouse/chick CNS development
6	Panina et al. 2024, P 38609948	Review/ functional (drug targeting)	**Competing**	SCP1 enzymatic class	Inhibitors target "small C-terminal domain phosphatase 1 (SCP1)"	Human glioblastoma

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7	UniProt O14595 GO (2026-09-08)	Database (IDA-backed)	**Competing**	True MF of ortholog	**GO:0008420 CTD phosphatase activity, IDA (direct assay)**; GO:0006470 dephosphorylation	Human
8	UniProt F7A4N8 GO (2026-09-08)	Database	**Refutes (absence)**	Any kinase annotation exists	**No GO terms; "kinase" appears 0**× in the record	Horse
9	InterPro F7A4N8 vs O14595 (2026-09-08)	Structural/ evolutionary	**Refutes/ qualifies**	Domain content of horse model	F7A4N8 = **no InterPro domains (HTTP 204)**; O14595 = FCP1/ HAD phosphatase domains, no kinase	Horse vs human
10	AlphaFold AF-F7A4N8-F1 v6 (2025-03-31)	Structural (predicted)	**Refutes/ qualifies**	Horse model has a folded catalytic domain	Mean pLDDT 46.2 (very low); 66% residues <50, 98% <70; no globular domain	Horse

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GO Curation Implications

- **GO:0016301 kinase activity — REMOVE / do not annotate.** No sequence, structural, or family evidence supports kinase activity for F7A4N8 or the CTDSP2 family. This is a misannotation (a phosphatase gene annotated with the mechanistically opposite activity). *(Lead — requires curator verification.)*
- **Better-supported MF (by orthology/family, requires the catalytic domain):** `GO:0004721 phosphoprotein phosphatase activity` (or more specific `GO:0008420 RNA polymerase II CTD heptapeptide repeat phosphatase activity` / `GO:0004725`-adjacent serine/threonine phosphatase), plus `GO:0046872 metal ion (Mg)`

`binding`. **Caveat:** these should be transferred to the horse gene with an ISS/orthology qualifier and a note that the current F7A4N8 protein model lacks the catalytic domain — the annotation belongs to the *gene*, not necessarily to this truncated protein isoform.

- **BP/CC (family-level, orthology):** BP `GO:0006470 protein dephosphorylation` / transcription regulation; CC `GO:0005634 nucleus`.
- Avoid "protein binding" as a terminal recommendation — the phosphatase MF is more informative and well-supported for the gene.

Mechanistic Scope

- **Immediate molecular function tested:** ATP-dependent phosphotransfer (kinase). **Not supported** — no ATP-binding/kinase fold in the sequence.
- **Established family function (direct gene-product activity):** Mg^{2+}-dependent HAD-family aspartate-nucleophile **phosphatase** that removes phosphate from phospho-Ser substrates (RNA Pol II CTD Ser5; REST degron phosphosites). This is a hydrolase reaction, not phosphotransfer.
- **Downstream/phenotypic (not the MF):** transcriptional regulation, REST stabilization, neuronal gene silencing, neurogenesis timing — consequences of phosphatase activity, not the activity itself.

Conflicts and Alternatives

- **Reaction-direction confusion (kinase vs phosphatase):** The most likely source of the erroneous "kinase activity" term. Both terms concern phosphate, but the gene is a phosphatase. No paralog of CTDSP2 is a kinase.
- **Truncated/aberrant gene model:** F7A4N8's C-terminus (res 85–174) is Ser/Arg-rich low-complexity and diverges from all mammalian CTDSP2 orthologs, which continue into a conserved catalytic domain. This looks like an incomplete or frameshifted Ensembl prediction rather than a genuine catalytic-domain-less isoform; it warrants caution before transferring *any* enzymatic MF to this specific protein record.

- **Isoform/orthology:** Identity is confirmed as horse CTDSP2 by gene name and VGNC/Ensembl xrefs and by the 96% N-terminal match to human CTDSP2 — but the protein model is not full-length.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Is F7A4N8 a real truncated isoform or a mis-prediction?	UniProt features + alignment show divergence at res ~84 and missing catalytic domain	Determines whether *this protein* has any enzymatic MF	Inspect Ensembl gene model / RNA-seq / a corrected full-length horse CTDSP2 transcript
Direct horse enzymatic assay	None found	No directly observed horse phosphatase/kinase data exists	In vitro pNPP/CTD-peptide phosphatase assay on full-length horse CTDSP2
Provenance of the kinase GO term	Not in supplied context	Whether it is IEA/ARBA carry-over vs. a specific claim	Trace annotation source & evidence code in the review

Discriminating Tests

1. **Retrieve full-length horse CTDSP2** (Ensembl/RefSeq) and re-scan for the DLDET motif and FCP1 domain — expected present in the true gene, absent in F7A4N8's current model.
2. **HMM/domain search (Pfam PF03031 "NIF"/FCP1-like) and structure (AlphaFold F7A4N8)** to confirm the catalytic core is absent from this model.
3. **Phosphatase activity assay** (para-nitrophenyl phosphate or phospho-CTD peptide) on recombinant full-length horse CTDSP2 — positive would confirm phosphatase; a kinase assay (ATP transfer) would be negative.

Curation Leads (require curator verification)

- **Action:** Remove/reject GO:0016301 "kinase activity" for CTDSP2 (F7A4N8) as a misannotation.
- **Replacement MF leads:** `GO:0004721 phosphoprotein phosphatase activity` / `GO:0008420 RNA Pol II CTD phosphatase activity`; add `GO:0046872 magnesium ion binding`; BP `GO:0006470 protein dephosphorylation`; CC `GO:0005634 nucleus` — all as **orthology (ISS)** with a note that the current F7A4N8 model lacks the catalytic domain.
- **Candidate references to verify:**

  P 30217818

  (SCP1 phosphatase kinetics + structure),

  P 17403776

  (SCP1 phosphatase in CNS development),

  P 38609948

  (SCP1 as a phosphatase drug target); database: UniProt O14595 (human CTDSP2, reviewed phosphatase).
- **Suggested question for curator:** Should enzymatic MF terms be attached to the F7A4N8 protein record at all, given it is a truncated model missing the catalytic domain — or annotated at the gene level only?

Limitations

- No directly observed *Equus caballus* biochemical data were found; family/ortholog transfer relies on human/mouse SCP paralogs. Motif and alignment analyses are computational. PubMed queries for "SCP2/CTDSP2" specifically returned no results; cited evidence is for the closely related paralog SCP1 and the family.
